Retention You Can Predict, Scope and Delete: a (μ, γ, T) Law and a Structural Deletion Guarantee for a Physics-Structured Memory Store

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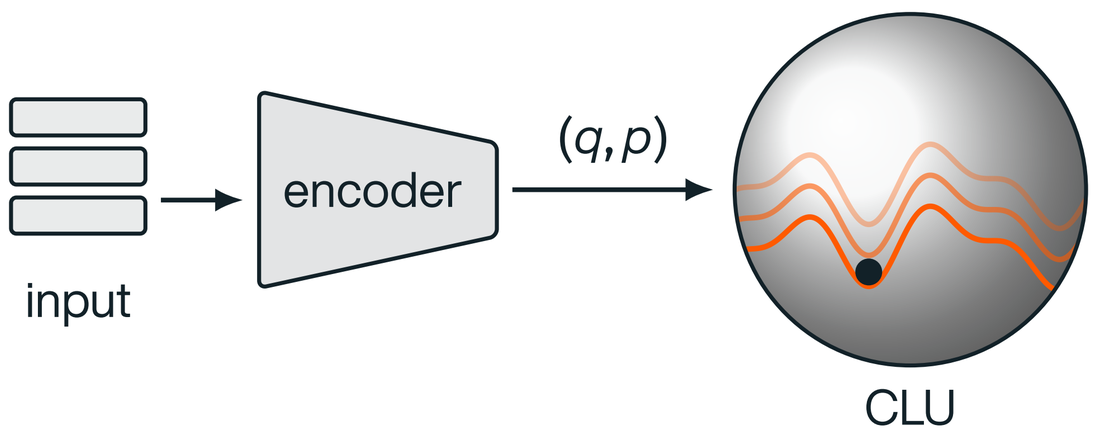
Forgetting in this store follows a computable law in spectral mass μ, damping γ and temperature T.

0.902
where forgetting is fastest, as a fraction of $2\varepsilon\mu$ (±0.003, learned units)

11
orders of magnitude in μ^2 on one curve, one measurement method

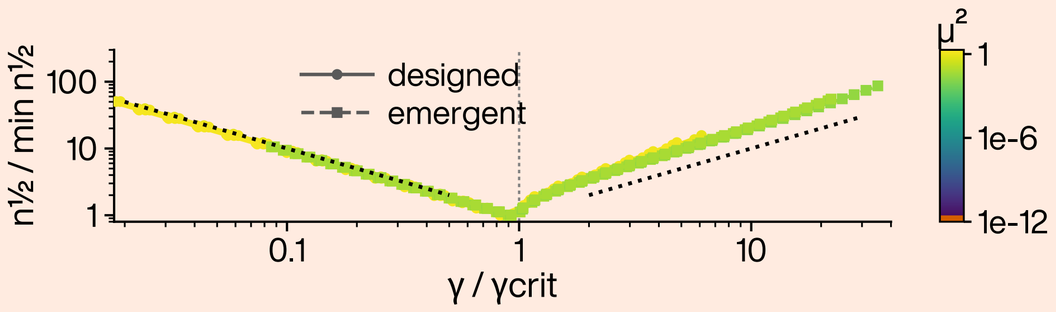
107.77×
longer retention inside a friction hole (±4.78, designed units)

1 The store is a CLU.



Spectral mass: μ^2 is the curvature of the potential along the stored direction (mass-weighted). Higher curvature, shorter memory. Half-life $n_{1/2}$: steps until a stored value fades to half. Designed: built-in symmetry. Emergent: learned.

2 A computable critical point.



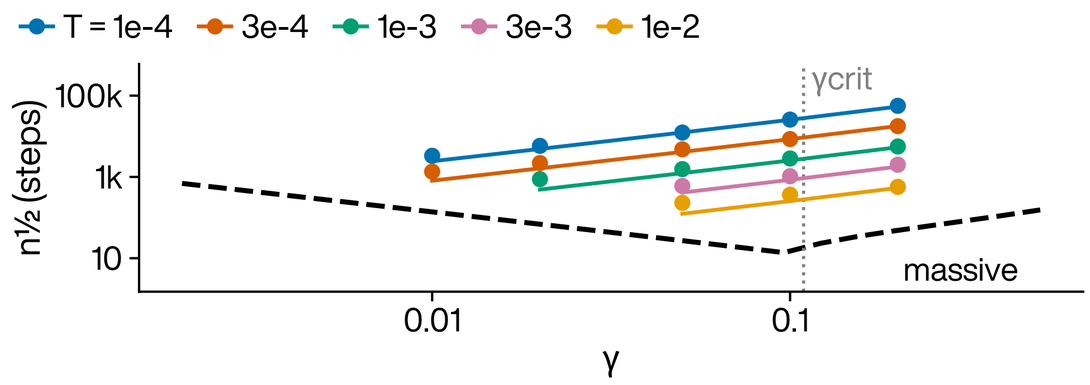
Half-life is shortest near $\gamma_{crit} \approx 2\varepsilon\mu$, for designed and learned units. The curve spans 11 orders of magnitude in μ^2; the low end is the resolution limit of the measurement. A second measurement method agrees to 0.35%.

3 Retention is predictable.

$$n_{1/2} \simeq \frac{2 \ln 2\, \gamma}{(2-\gamma)\,\varepsilon^2\mu^2} \propto \frac{1}{\mu^2} \quad (\gamma > \gamma_{crit}, \text{ overdamped side})$$

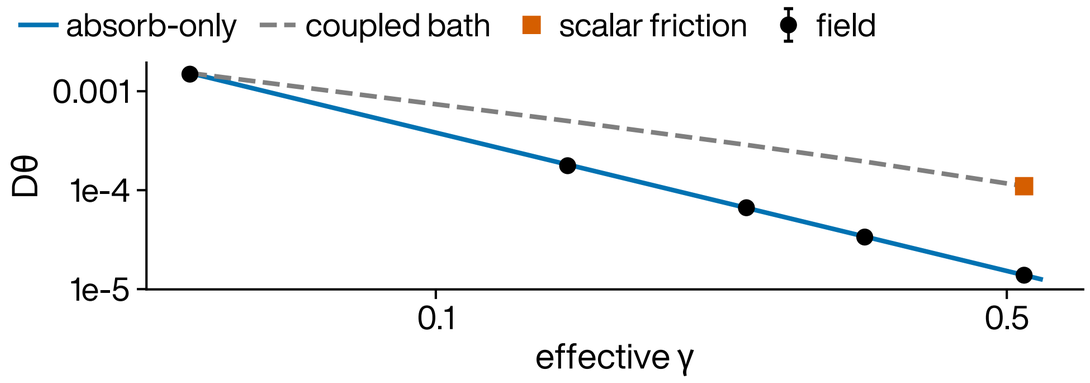
Next: larger scales, task-level tests and local temperature fields.

4 Friction preserves, T erases.



More friction, longer retention; more temperature, shorter. The heavy mode does the opposite.

5 A friction hole is a memory vault.



Values last 107.77 ± 4.78× longer inside the hole than outside; uniform friction at the same damping gives 13.28×.

6 Deletion is exact for the store itself.

- After deletion the store is the same as if the item was never written, below and above full capacity.
- Needs: designed store, budget, no leakage, priority eviction.
- Not covered: a learned encoder; certified unlearning. Trade-off: lifetime depends on amplitude.
- Scope: early results on a designed testbed, not a deployed agent memory.

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